

GREETINGS

Dear members of the commission offer you a term paper on the topic: game "Chess".

ADJECTIVE

My goal for the course work was to develop a chess game in c++ using the sfml library

DEVELOPMENT TOOLS

I used the c++ programming language together with the sfml multimedia library. Also, the program LibreSprite was chosen as a graphic editor. It was used to draw all the textures used in the game.

To run the program, I used the GCC compiler

ARCHITECTURE AND GAME CLASSES

The main class in the program is the Game class. It is responsible for all the logic of the program: changing the state of the game, the turn queue, rendering objects, checking the validity of the move, etc.

The parent class for all figures is the Figure class. It describes all the methods and parameters that are used for each figure class, except for the method for finding moves, this method is considered a polymorphism.

The Board class is responsible for displaying the board and simulating the game in a two-dimensional array. In this class, you can access the two-dimensional array responsible for the position of the pieces only using getters and setters, which is a good example of encapsulation.

The Hint class is responsible for displaying the possibility of a move.

The figure shows a general UML diagram for the game "Chess". It shows the relationships between each class, an example of inheritance, and polymorphism

RENDERING

The game contains a lot of objects and game elements, which need to render. Also, during interaction, some game objects could be deleted from game process. Object Board renders while program is running, and deletes when program close. Other objects are dynamically created and removed during program execution.

GAMEPLAY TESTING

The pictures show general examples of chess moves, such as: castling, the first move for a pawn, moving a pawn to the end of the playing field, an example of a draw and a checkmate.

CONCLUSION

In the further development of the software product, it is planned to add the following functions and code modifications:

As a result of the course work, a single-player game “Chess” was designed and developed using the SFML library for visualization and control of the gameplay. At all stages of the project, the principles of object-oriented programming were applied: encapsulation, inheritance, polymorphism Particular attention was paid to the correct processing of game events, changing game states and implementing special chess mechanics such as rocking, filtering and processing moves for a piece, checkmate, checkmate and draw.